MTH 201 Chapter 5: Parametric/Polar Curves & Conics

5.1 Parametric Equations

Not all curves pass the vertical line test. It is impossible to describe such curves in the form $y = f(x)$. The idea in this section is to express such curves in the xy-plane with parametric equations.

Terminology:

__**Example 1:**__ (a) Sketch the curve given by the parametric equations: $x = t^2 - 2t \quad y = t - 1$.

(b) Describe this curve with a single equation which does not have the parameter t in it.

Example 2: One nice application of parametric equations is that they can be used to model the movement of a particle that traverses some curve in the plane (in some applications we view t as the time variable). Draw the path traversed by a particle whose movement is modeled by the parametric equations:

$$x = t^2 - 2t \quad y = t - 1 \text{ where } -2 \le t \le 2 \ .$$

Note: Many graphs that cannot be graphed in xy-coordinates can be expressed as parametric equations.

Example 3: Graph $x = \sin(y) \quad -2\pi \le y \le 2\pi$ parametrically.

Example 4 (you try): (unrelated parts)

(a) Sketch the curve with parametric equations: $x = \cos(t) \quad y = \cos^2(t)$

(b) Find parametric equations for the curve determined by the equation: $x = y^4 - 3y^2$

(c) Find parametric equations which describe the movement of a particle which traverses a circle of radius 4 exactly one time in the counterclockwise direction.

Calculus with Parametric Curves

Motivation: Suppose f and g are differentiable functions and we want to find the tangent line at a point on the curve $x = f(t)$, $y = g(t)$. If we suppose that y is a differentiable function of x. Then,

Theorem:

Note: The proof of this theorem is beyond the scope of the course.

Note:

<u>Example 5:</u> Suppose that a curve C is defined by the parametric equations $x = t^2$, $y = t^3 - 9t$.

(a) Find the equation of both tangent lines to C at $(9,0)$.

(b) Find the point(s) on C where a tangent is horizontal or vertical.

(c) For what values of t is the curve concave upward/downward.

Example 6 (you try):

For which values of t is the curve given by $x = t^2 + 1,\ y = e^t - 1$ concave upward/downward.

Using the ideas from chapter 2 & 3 we can find ways to compute areas beneath/between parametric curves and derive formulas for arc length and surface area in the setting of parametric equations. Our textbook only discusses arc length. I will include extra (optional) notes on the other topics mentioned above.

Theorem:

Proof: Shown in the textbook.

Example 7: Find the exact length of the curve given by $x = e^t + e^{-t}$ $\quad y = 5 - 2t$ where $0 \le t \le 3$.

Example 8 (you try): Set up an integral that represents the length of the curve given by $x = t^2 - t$ $\quad y = t^4$ where $1 \le t \le 4$. Then, use your calculator to find the approximate length to four decimal places.